

THE GALOIS GROUP OF $x(x-1)\cdots(x-p+1)+1$

CLAUDE (ANTHROPIC)

ABSTRACT. Let p be an odd prime and $f_p(x) = x(x-1)\cdots(x-p+1)+1$. We prove a general certificate: a monic irreducible of prime degree p whose Frobenius at an odd unramified prime q is both odd (Stickelberger) and fixed-point-free has Galois group S_p . The group-theoretic input is that every transitive nonsolvable subgroup of S_p is S_p or lies in A_p (CFSG only through Guralnick). For f_p the fixed-point-free condition is automatic at every $q \leq p$, so a single Kronecker symbol $(\text{disc } f_p/q) = -1$ with $q < p$ yields $\text{Gal}(f_p/\mathbb{Q}) = S_p$. This verifies the identification for every odd prime $p < 10^7$, and for all primes in six residue classes modulo 36 (Dirichlet density $\frac{1}{2}$). A second certificate uses no classification at all: if $f \bmod q$ has an isolated irreducible factor of prime degree $\ell \in [3, p-3]$ and Frob_q is odd, then Dedekind and Jordan alone give S_p . A witness $q \leq 61$ exists for every prime $p < 1500$. For the two-parameter family $(x)_p + c$ both certificates persist, and a critical-window theorem identifies S_p for every p and every c with $\mu_{m-1} < |c| < \mu_{m-2}$, by complex conjugation alone; $c = 1$ is the hard case precisely because it lies below every critical magnitude. The polynomial is totally real for $p \geq 5$, so conjugation is not a transposition; Newton polygons are flat; φ is Morse in the sense of Serre and $\text{Gal}(\varphi - t/\mathbb{Q}(t)) = S_p$. The remaining obstruction to a proof for every p is whether the periodic good classes of $(\text{disc } f_p/q)$, as q varies, cover every prime.

1. THE POLYNOMIAL

Let p be an odd prime, $\varphi(x) = (x)_p = x(x-1)\cdots(x-p+1)$, and

$$f_p(x) = \varphi(x) + 1 \in \mathbb{Z}[x].$$

Reduction modulo p gives $f_p \equiv x^p - x + 1 \pmod{p}$, which is irreducible over $\mathbb{F}_p$ (Artin–Schreier). Hence f_p is irreducible over $\mathbb{Q}$, and $G = \text{Gal}(f_p/\mathbb{Q}) \leq S_p$ is transitive of prime degree.

The polynomial belongs to the family raised by Schur in 1908 [11]: is $(x-a_1)\cdots(x-a_n) \pm 1$ irreducible for distinct integers a_i ? Westlund and Flügel answered it the following year [20]: the -1 form is always irreducible, and the $+1$ form is irreducible unless it is, after a translation, $x(x-2)+1$ or $x(x-1)(x-2)(x-3)+1$. For the consecutive family this says that $(x)_n+1$ is irreducible for every $n \neq 4$; see [13] for a modern account and the later generalisations. Irreducibility is therefore classical in every degree, and the Artin–Schreier argument above is quoted only to keep the prime-degree case self-contained. What is at issue here is the Galois group.

Proposition 1. *If $q \leq p$ is prime and $a \in \mathbb{F}_q$, then $f_p(a) = 1$. In particular f_p has no root in $\mathbb{F}_q$.*

Proof. The $p \geq q$ consecutive factors of $\varphi(a)$ cover every residue in $\mathbb{F}_q$, so $\varphi(a) = 0$ and $f_p(a) = 1$. $\square$

Thus Frob_q , for any prime $q \leq p$ not dividing $\text{disc } f_p$, is fixed-point-free (Dedekind). At $q = p$ the reduction $f_p \bmod p$ is irreducible, hence separable, so $p \nmid \text{disc } f_p$, and Frob_p is a p -cycle.

2. A GENERAL CERTIFICATE

Theorem 2. *Let $f \in \mathbb{Z}[x]$ be monic irreducible of degree p an odd prime, and let q be an odd prime with $q \nmid \text{disc } f$ such that*

- (1) *f has no root in $\mathbb{F}_q$, and*
- (2) *$\left(\frac{\text{disc } f}{q}\right) = -1$.*

Then $\text{Gal}(f/\mathbb{Q}) = S_p$.

The second hypothesis is Stickelberger's parity theorem: $\text{sgn}(\text{Frob}_q) = (\text{disc } f/q) \pmod{2}$ [3, 2].

Lemma 3. *Let p be an odd prime and $G \leq S_p$ transitive and nonsolvable. Then $G = S_p$ or $G \leq A_p$.*

Equivalently, the only transitive subgroups of S_p containing an odd permutation are S_p and certain subgroups of $\text{AGL}(1, p)$.

Proof of Theorem 2, given Lemma 3. Let $G = \text{Gal}(f/\mathbb{Q}) \leq S_p$, transitive, and $\sigma = \text{Frob}_q$. By Dedekind, σ has no fixed point; by Stickelberger, σ is odd. If G were solvable, Galois's theorem would embed it in $\text{AGL}(1, p) = \{x \mapsto ax + b\}$. An affine map with $a \neq 1$ fixes $b/(1 - a)$, so the only fixed-point-free non-identity elements are translations, i.e. p -cycles, which are even (p odd). This contradicts σ . Thus G is nonsolvable, so Lemma 3 gives $G = S_p$ or $G \leq A_p$; oddness of σ excludes the latter. $\square$

Remark 4 (Hypothesis (1) can be weakened). Only the fixed-point *count* matters, and only through its being $\neq 1$. A non-identity element $x \mapsto ax + b$ of $\text{AGL}(1, p)$ with $a \neq 1$ of order d has type $(1, d^{(p-1)/d})$ and sign $(-1)^{(d-1)(p-1)/d}$, so it is odd exactly when d is even and $(p-1)/d$ is odd — and then it fixes exactly one point.

For the certificate one needs less than this. The elements with $a = 1$ are the translations, whose order divides p and is therefore 1 or p ; both are odd, so a translation is an even permutation, no sign formula required. Hence an odd element must have $a \neq 1$, and then it fixes exactly the point $b/(1 - a)$. So every odd element of $\text{AGL}(1, p)$ has precisely one fixed point, and hypothesis (1) of Theorem 2 may be replaced by: *f has a number of roots in $\mathbb{F}_q$ different from 1.*¹ For a polynomial with Galois group S_p this raises the density of usable q from $\approx e^{-1}$ to $\approx 1 - e^{-1}$. It changes nothing for f_p , where Proposition 1 already gives zero roots at every $q \leq p$.

Remark 5 (Prime degree is essential). Not merely convenient: $C_4 = \langle (1234) \rangle \leq S_4$ is transitive and its generator is an odd derangement, so the conclusion

¹This paragraph is formalised in Lean 4 against `mathlib`, in `formal/Formal/AGLCycleTypes.lean`: `existsUnique_fixed_of_sign_eq_neg_one` and its contrapositive `sign_eq_one_of_not_existsUnique_fixed`. The development contains no `sorry`, and the results depend only on the three standard axioms `propext`, `Classical.choice` and `Quot.sound`. It is the argument just given, in the form that avoids the sign formula.

of Theorem 2 fails at the first composite degree. The proof uses primality twice — Galois’s theorem for the solvable case, and Burnside’s for the rest — and both fail for composite n . By Westlund–Flügel the polynomials $(x)_n + 1$ are irreducible for every $n \neq 4$, so irreducibility is no longer the obstacle to a composite-degree statement; transitivity is not enough, and primitivity would have to be established separately.

Remark 6 (Sharpness). Neither hypothesis on σ can be dropped. A primitive root in $\text{AGL}(1, p)$ gives type $(1, p-1)$, sign $(-1)^{p-2} = -1$; translations are fixed-point-free and even. Likewise A_p contains p -cycles. Only the conjunction “odd and fixed-point-free” is impossible outside S_p .

We first isolate the group-theoretic fact needed in the $\text{PSL}_2(11)$ case.

Lemma 7. $\text{PGL}_2(11)$ has no subgroup of index 11.

Proof. Suppose $H \leq \text{PGL}_2(11)$ with $|H| = 120$. Since $120 \nmid 660$, one has $H \not\leq \text{PSL}_2(11)$. The image of H in $\text{PGL}_2(11)/\text{PSL}_2(11) \cong C_2$ is therefore nontrivial, so $H/(H \cap \text{PSL}_2(11)) \hookrightarrow C_2$ has order 2 and $H_0 = H \cap \text{PSL}_2(11)$ has order 60. By Dickson’s list [6, Hauptsatz II.8.27], the only subgroups of order 60 in $\text{PSL}_2(11)$ are isomorphic to A_5 . Then $H \leq N := N_{\text{PGL}_2(11)}(H_0)$ and $N \cap \text{PSL} = N_{\text{PSL}}(A_5) = A_5$ (A_5 maximal and non-normal), so $|N| \in \{60, 120\}$. If $|N| = 120$, the conjugation map $N \rightarrow \text{Aut}(A_5)$ has kernel $C_N(A_5)$ meeting A_5 trivially, hence either

- $N = A_5 \times C_2$: then $A_5 \leq C_{\text{PGL}_2(11)}(z)$ for an involution z . Every involution in $\text{PGL}_2(11)$ fixes 0 or 2 points of $\mathbb{P}^1(\mathbb{F}_{11})$, so its centralizer lies in a torus normalizer of order $2(q \mp 1) \in \{20, 24\} < 60$.
- $N \cong S_5$: then N contains F_{20} , in which an element induces an automorphism of order 4 of C_5 . But a Sylow C_5 of $\text{PGL}_2(11)$ lies in the split torus and fixes exactly two points, so $N(C_5)$ permutes that pair with kernel the diagonal torus C_{10} : order at most 20, image at most C_2 in $\text{Aut}(C_5) \cong C_4$.

Thus $|N| = 60 < 120 = |H|$, a contradiction. $\square$

Proof of Lemma 3. By Burnside, a transitive group of prime degree is solvable (hence affine) or 2-transitive. So G is 2-transitive, hence primitive. The socle of a 2-transitive group is regular elementary abelian or nonabelian simple; the abelian case is affine, excluded. Thus G is almost simple: $S \leq G \leq \text{Aut}(S)$ with S nonabelian simple, transitive, and $[S : S_\omega] = p$. By Guralnick [5] (cf. Fengler [4, Cor. 2.39]), (S, S_ω) is one of:

- (1) $S = A_p$, $S_\omega = A_{p-1}$;
- (2) $S = \text{PSL}_d(q_0)$, $p = (q_0^d - 1)/(q_0 - 1)$, S_ω a point or hyperplane stabilizer (d prime);
- (3) $S = \text{PSL}_2(11)$, $S_\omega = A_5$, $p = 11$;
- (4) $S = M_{11}$ ($p = 11$) or $S = M_{23}$ ($p = 23$).

(The remaining case of Guralnick’s theorem, $S = \text{PSU}_4(2)$ with a subgroup of index 27 [4, Thm. 2.38], does not occur: $27 = 3^3$ is a prime power but not a prime.)

The restriction $\text{sgn}|_S$ has kernel of index at most 2; a nonabelian simple group has no subgroup of index 2. So $S \leq A_p$ and sgn factors through $G/S \hookrightarrow \text{Out}(S)$.

Case A_p . Then $G \in \{A_p, S_p\}$.

Case M_{11}, M_{23} . $\text{Out}(M_{11}) = \text{Out}(M_{23}) = 1$ [10], so $G = S \leq A_p$.

Case $\text{PSL}_2(11)$. Here $\text{Aut}(\text{PSL}_2(11)) = \text{PGL}_2(11)$, so $S < G$ would force $G = \text{PGL}_2(11)$, whose transitive action on the 11 points would require a subgroup of index 11, contradicting Lemma 7. Hence $G = \text{PSL}_2(11) \leq A_{11}$.

Case $\text{PSL}_d(q_0)$, $q_0 = r^e$. Let $\Omega = \mathbb{P}^{d-1}(\mathbb{F}_{q_0})$ and take S_ω to be a point stabilizer (composing with the graph automorphism if needed). Distinct points of a 2-transitive group have distinct stabilizers, so if $c \in C_{\text{Sym}(\Omega)}(S)$ then $S_{c(\omega)} = cS_\omega c^{-1} = S_\omega$, hence $c = 1$; the trivial centralizer is used in (c) below. Separately, since $S \trianglelefteq G$, we have $G \leq N_{\text{Sym}(\Omega)}(S)$, and every $g \in G$ satisfies $gS_\omega g^{-1} = S_{g\omega}$, so g preserves the S -class of the point stabilizer.

(a) *No diagonal part.* As d is prime, $d \mid q_0 - 1$ would give $p \equiv d \equiv 0 \pmod{d}$, hence $p = d < 2^d - 1 \leq p$, absurd. Thus $\text{PGL}_d(q_0) = \text{PSL}_d(q_0)$.

(b) *No graph part, $d \geq 3$.* The graph automorphism swaps the point- and hyperplane-stabilizer classes [10, §3.3], while field automorphisms preserve the class of a point stabilizer. Class preservation therefore forces $G \leq \text{PTL}_d(q_0)$ and $G/S \leq \langle \varphi_r \rangle$.

(c) *The field part is even.* Any $g \in G$ inducing φ_r^k equals $s \cdot \varphi_r^k$ in $\text{Sym}(\Omega)$ with $s \in S \leq A_p$, so $\text{sgn}(g) = \text{sgn}(\varphi_r)^k$. If e is odd, φ_r has odd order, all cycles have odd length, and φ_r is even. This covers all $d \geq 3$: if e were even, $s_0 = r^{e/2}$ would give

$$p = \frac{s_0^d - 1}{s_0 - 1} \cdot \frac{s_0^d + 1}{s_0 + 1}$$

with both factors integers > 1 (d odd). For $d = 2$, $p = q_0 + 1$ prime forces $q_0 = 2^e$ with $e = 2^j$ (Fermat). On $\mathbb{P}^1(\mathbb{F}_{2^e})$, φ_2 fixes $\{0, 1, \infty\}$ and every nontrivial orbit has 2-power length ≥ 2 , hence is a cycle of even length and an odd permutation, so $\text{sgn}(\varphi_2) = (-1)^t$ with t the number of nontrivial orbits. The number of orbits of length 2^i is the number of monic irreducibles of degree 2^i over $\mathbb{F}_2$,

$$n_i = \frac{2^{2^i} - 2^{2^{i-1}}}{2^i}, \quad v_2(n_i) = 2^{i-1} - i.$$

Thus $n_1 = 1$ and $n_2 = 3$ are odd, while n_i is even for $i \geq 3$. For $j \geq 2$, $t \equiv n_1 + n_2 \equiv 0 \pmod{2}$ and φ_2 is even. For $j = 1$, $\text{PTL}_2(4) \cong S_5 = S_p$.

In all cases $G = S_p$ or $G \leq A_p$. $\square$

Remark 8 (Which p need the classification at all). Case (2) of the proof occurs only when $p = (q_0^d - 1)/(q_0 - 1)$ for some prime power q_0 and prime d — a *projective prime* in the terminology of Jones and Zvonkin [16], who observe that the classification of transitive groups of prime degree is complete except for the open question of which integers of this shape are prime, and give evidence that infinitely many are. If p is not a projective prime and $p \notin \{11, 23\}$, the only transitive subgroups of S_p are the subgroups of $\text{AGL}(1, p)$, A_p and S_p . For such p , Lemma 3 needs only Burnside and the affine cycle-type computation, and neither Guralnick nor Lemma 7 is used. The projective primes below 100 are 3, 5, 7, 13, 17, 31 and 73 (those with $d = 2$ are exactly the Fermat primes, as in the proof above); together with 11 and 23 they are the only $p < 100$ for which the classification is invoked.

CFSG enters only through Guralnick's theorem. The facts $\text{Out}(M_{11}) = \text{Out}(M_{23}) = 1$ and non-conjugacy of the two parabolic classes for $d \geq 3$ are in [10]; maximality of A_5 in $\text{PSL}_2(11)$ is in Dickson's list [6, Hauptsatz II.8.27]. The involution-centralizer bound and the absence of F_{20} in $\text{PGL}_2(11)$, and the signs of φ_2 on $\mathbb{P}^1(\mathbb{F}_{2^e})$ for $e = 2, 4, 8, 16$, are finite checks. (If an order-4 element conjugated an element of order 5 to its cube, its inverse would conjugate it to its square, so testing $bab^{-1} = a^2$ over all order-4 elements covers both generators of $\text{Aut}(C_5)$.)

3. THE CERTIFICATE FOR f_p

Corollary 9. *Let p be an odd prime. If some odd prime $q < p$ satisfies $(\frac{\text{disc } f_p}{q}) = -1$, then $\text{Gal}(f_p/\mathbb{Q}) = S_p$.*

Proof. Hypothesis (1) of Theorem 2 holds for every prime $q \leq p$ by Proposition 1, so Theorem 2 applies. $\square$

The prime $q = p$ can never serve: Frob_p is an even p -cycle, equivalently $\text{disc } f_p \equiv (-1)^{(p+1)/2} \pmod{p}$ is always a residue. Evaluating the symbol at a candidate $q < p$ is a single resultant over $\mathbb{F}_q$. For such q ,

$$f_p \equiv (x^q - x)^{\lfloor p/q \rfloor} \prod_{k < p \bmod q} (x - k) + 1 \pmod{q},$$

and $\text{disc } f_p \bmod q$ is the discriminant of this reduction; the symbol is its quadratic character. A search over odd primes $q < p$ finds a witness $q \leq 73$ for every prime $7 \leq p < 10^7$ (largest witness $q = 73$ at $p = 9683099$; $q = 3$ succeeds for 50.003% of the 664577 primes, matching the density $\frac{1}{2}$ of the classes in Proposition 24 to five significant figures). For $p = 5$ the only candidate is $q = 3$, and $(38569/3) = +1$, so the certificate has no witness; S_5 is settled instead by the type-(2, 3) Frobenius at $q = 19$. For $p = 3$, $\text{disc } f_3 = -23$ is not a square, so $G = S_3$.²

4. A CFSG-FREE CERTIFICATE

Theorem 2 buys its conclusion with Guralnick's theorem. For f_p there is a second, entirely classical certificate, which uses no classification at all. It is the standard Jordan–Dedekind test of computational Galois theory [18, 15, 17]; what is new here is only that it fires for this family, and that recording the witness makes the identification independent of §7 and of CFSG.

Theorem 10. *Let $f \in \mathbb{Z}[x]$ be monic irreducible of prime degree $p \geq 7$, and let q be a prime such that $f \bmod q$ is separable, with irreducible factors of degrees $\ell_1, \dots, \ell_k$. Suppose*

- (1) *some prime $\ell \in [3, p-3]$ occurs exactly once among the ℓ_i and is coprime to every other ℓ_j , and*

²The complete witness list for $7 \leq p < 10^7$, the $p \leq 97$ discriminant data, the $q = 3$ and $q = 5$ class tables of §7, the measured ε_q for odd $q \leq 199$, and the verification programs are included with this note (`ancillary/` for data, `tools/` for the checkers, one per computational claim). Each row (p, q) of the witness list is verifiable independently of the search that produced it, by one resultant and one Legendre symbol over $\mathbb{F}_q$.

(2) $p - k$ is odd.

Then $\text{Gal}(f/\mathbb{Q}) = S_p$.

Lemma 11 (Isolation; [15, Comments 1]). *Let σ have cycle type $(\ell_1, \dots, \ell_k)$ and suppose $\ell = \ell_i$ is coprime to $N = \text{lcm}_{j \neq i} \ell_j$. Then σ^N is an ℓ -cycle.*

Proof. Raising to the power N kills every cycle of length ℓ_j , $j \neq i$. On the remaining cycle, $\gcd(\ell, N) = 1$, so σ^N restricts to a generator of that cycle's cyclic group; it is an ℓ -cycle with $p - \ell$ fixed points. $\square$

Proof of Theorem 10. Let $G = \text{Gal}(f/\mathbb{Q}) \leq S_p$, transitive because f is irreducible, hence primitive: a block system has blocks of size dividing p . Separability of $f \bmod q$ gives $q \nmid \text{disc } f$, so q does not divide the index $[\mathcal{O}_K : \mathbb{Z}[\alpha]]$ and Dedekind's theorem [12] applies: $\sigma = \text{Frob}_q$ has cycle type $(\ell_1, \dots, \ell_k)$. By Lemma 11, σ^N is an ℓ -cycle with ℓ prime and $3 \leq \ell \leq p - 3$, so Jordan's theorem [19, Thm. 13.9] gives $A_p \leq G$. Finally $\text{sgn } \sigma = \prod_i (-1)^{\ell_i - 1} = (-1)^{p-k} = -1$, so $G \not\leq A_p$ and $G = S_p$. $\square$

The two certificates are complementary, and they use disjoint hypotheses on Frob_q . Theorem 2 needs σ to have no fixed point, which for f_p is automatic at every $q \leq p$ but says nothing about the rest of the cycle type; Theorem 10 needs an isolated cycle of prime length $\leq p - 3$, a condition on the cycle type that is unrelated to the fixed-point count, and the two are met together only by accident. Both require σ odd, and neither can be replaced by the other.

Remark 12 (Affine types never certify). A non-identity element of $\text{AGL}(1, p)$ is a p -cycle or has type $(1, d^{(p-1)/d})$ with $d = \text{ord}(a) > 1$. In the first case $\ell = p > p - 3$; in the second, a cycle of length d is isolated only if $(p-1)/d = 1$, i.e. $d = p - 1$, which is even and exceeds $p - 3$. So no affine element satisfies hypothesis (1), consistently with Jordan's theorem.

Remark 13 (Why this does not settle every p). Whether a witness exists for every p is the same kind of question as the covering problem of §8, and it is circular in the same way: Chebotarev predicts the density of good q only once G is known. What can be said is that the prediction is favourable. By Unger [18, Thm. 5], the proportion of elements of S_n that do not power to a cycle of prime length is $O(\log \log n / \log n)$; so if $G = S_p$, the density of usable q tends to 1, against the density $\varepsilon_q \approx \frac{1}{2}$ per prime available to Corollary 9. Per prime tried, the Jordan test is the better instrument, and provably so.

Hypothesis (1) is checked by a distinct-degree factorisation of $f_p \bmod q$, which returns the multiset $\{\ell_i\}$ without producing the factors. For every prime $7 \leq p < 1500$ a witness exists with $q \leq 61$; the largest least-witness is $q = 61$, attained at $p = 1301$, and $q = 3$ already succeeds for 80 of the 236 primes in the range, $q \leq 7$ for 169 of them. The witnesses are listed in `ancillary/jordan_witnesses.txt`, one row $(p, q, \ell, \{\ell_i\})$ per prime, and each row is re-derived from (p, q) alone by `tools/verify_jordan.py`. For those p , the identification $\text{Gal}(f_p/\mathbb{Q}) = S_p$ rests on Dedekind and Jordan alone.

5. CLOSED ROUTES

Proposition 14 (Totally real). *For every prime $p \geq 5$, f_p has p real roots. (For $p = 3$ there is one real root.)*

Proof. Write $x_k = k - \frac{1}{2}$ for $0 \leq k \leq p$. The sign of $\varphi(x_k)$ is $(-1)^{p-k}$, and

$$\frac{|\varphi(x_{k+1})|}{|\varphi(x_k)|} = \frac{k + \frac{1}{2}}{p - k - \frac{1}{2}}.$$

The sequence $|\varphi(x_k)|$ is therefore strictly unimodal with minimum at the centre $m = (p-1)/2$,

$$\min_k |\varphi(x_k)| = \frac{(2m-1)!! (2m+1)!!}{2^p}.$$

This equals $45/32 > 1$ at $p = 5$ and thereafter grows by the factor $(2m+1)(2m+3)/4$. Hence $f_p = \varphi + 1$ has the sign of φ at all $p+1$ half-integers, so p sign changes and p real roots. At $p = 3$ the minimum is $3/8 < 1$. $\square$

Complex conjugation is therefore not a transposition for $p \geq 5$, and Conrad's criterion (irreducible of prime degree, exactly two non-real roots) does not apply.

Proposition 15. *The polynomial φ is a Morse function in the sense of Serre [9, §4.4]: the zeros of φ' are simple and the finite critical values are pairwise distinct. Consequently $\text{Gal}(\varphi - t/\mathbb{Q}(t)) = S_p$ [9, Thm. 4.4.5].*

Proof. On each interval $(k, k+1)$, $k = 0, \dots, p-2$, the logarithmic derivative $H(x) = \sum_{j=0}^{p-1} (x-j)^{-1}$ is strictly decreasing ($H' = -\sum (x-j)^{-2} < 0$), so there is a unique simple zero c_k . Thus $\varphi''(c_k) = \varphi(c_k)H'(c_k) \neq 0$. (The point ∞ is a totally ramified critical point of multiplicity p , excluded from Serre's definition.) The involution $x \mapsto (p-1) - x$ sends c_k to c_{p-2-k} and φ to $-\varphi$.

Write $\mu_k = \max_{(k, k+1)} |\varphi|$ and $m = (p-1)/2$. For $x \in (k, k+1)$ one has $|\varphi(x+1)/\varphi(x)| = (x+1)/(p-1-x)$, which is < 1 throughout $(k, k+1)$ whenever $k \leq m-2$. Thus $|\varphi(x+1)| < |\varphi(x)|$ pointwise on that interval, and taking maxima yields $\mu_{k+1} < \mu_k$ for $k = 0, \dots, m-2$. The central pair has $\mu_{m-1} = \mu_m$ by the involution, so equal magnitudes occur only for mirror pairs $c_k \leftrightarrow c_{p-2-k}$. Those indices have opposite parity (p odd), hence opposite signs $(-1)^{p-1-k}$; since no critical value vanishes (the roots of φ are simple), the two members of a mirror pair are distinct. Critical values of different magnitude are distinct a fortiori, so all finite critical values are pairwise distinct. $\square$

(The specialisation $t = -1$ can still be exceptional: $(x)_4 + 1 = (x^2 - 3x + 1)^2$. Hilbert irreducibility of the family does not decide this single value.)

Newton polygons of f_p are flat: $f_p(0) = \dots = f_p(p-1) = 1$. The same holds for every translate $f_p(x+a)$ with $0 \leq a < p$, which is what excludes the method. Flatness is precisely the obstruction to the Coleman–Hajir criterion [14], which reads a cycle of prime-power length off a Newton slope and then applies Jordan's theorem; that is how the Galois groups of the generalized Laguerre polynomials are shown to contain A_n . Theorem 10 reaches the

same conclusion for f_p by reading cycle types off Dedekind instead. Hilbert–Grunwald and effective HIT prescribe or bound *some* specialisation of $\varphi - t$, never the fixed $t = -1$.

The arithmetic Morse criterion of [9, Remark 2, pp. 42–43] (at most one double root modulo every prime; inertia generates Gal by Minkowski; the worked example $x^n - x - 1$, following Nart–Vila [7] and Osada [8]) applies to f_p itself once $\text{disc } f_p$ is squarefree. Direct factorisation gives this for $p \leq 7$; already at $p = 97$ the discriminant has 12642 digits. Among $p \leq 97$, every ramified prime $\ell \leq 10^6$ has $v_\ell(\text{disc } f_p) = 1$. A single such ℓ produces an inertia transposition and, with the mod- p p -cycle, yields S_p ; this fires for 18 of the 24 primes $p \leq 97$.

6. THE FAMILY $(x)_p + c$

Nothing in §§2–4 uses the constant 1. Writing

$$f_{p,c}(x) = \varphi(x) + c = x(x-1) \cdots (x-p+1) + c, \quad c \in \mathbb{Z} \setminus \{0\},$$

the certificates apply verbatim, and the two-parameter family shows what is special about $c = 1$: it is the only regime in which the polynomial is totally real, and total reality is exactly what closes the elementary route.

Proposition 16. *Let p be an odd prime and $c \in \mathbb{Z}$ with $p \nmid c$.*

- (1) *$f_{p,c}$ is irreducible over $\mathbb{Q}$, and Frob_p is a p -cycle.*
- (2) *If $q \leq p$ is prime and $q \nmid c$, then $f_{p,c}$ has no root in $\mathbb{F}_q$.*

Consequently, if some odd prime $q < p$ with $q \nmid c$ satisfies $(\frac{\text{disc } f_{p,c}}{q}) = -1$, then $\text{Gal}(f_{p,c}/\mathbb{Q}) = S_p$.

Proof. (1) In $\mathbb{F}_p[x]$ one has $\varphi = \prod_{k \in \mathbb{F}_p} (x - k) = x^p - x$, so $f_{p,c} \equiv x^p - x + c \pmod{p}$. By Artin–Schreier, $x^p - x - a$ is irreducible over $\mathbb{F}_p$ exactly when $a \notin \wp(\mathbb{F}_p) = \{0\}$; here $a = -c \neq 0$. The reduction is irreducible, hence separable, so $p \nmid \text{disc } f_{p,c}$ and Frob_p is a p -cycle. (2) The $p \geq q$ consecutive factors of $\varphi(a)$ run over every residue in $\mathbb{F}_q$, so $\varphi(a) = 0$ and $f_{p,c}(a) = c \neq 0$ in $\mathbb{F}_q$. The last statement is Theorem 2 applied with hypothesis (1) supplied by (2). $\square$

The periodicity calculus of §7 also carries over: in (1) each factor $g(\beta)^m B_r(\beta) + 1$ becomes $g(\beta)^m B_r(\beta) + c$, and the period bound $\Pi(q)$ is unchanged, since it depends only on $m \bmod q$ and on the multiplicative orders of the $g(\beta)$. Only the good classes move with c .

We now show that for $|c|$ in one explicit window the identification is elementary — no Frobenius, no classification, no computation.

Lemma 17 (Root count). *Let $p \geq 5$ be prime, let $c_0 < \cdots < c_{p-2}$ be the critical points of φ from Proposition 15, and put $\mu_k = |\varphi(c_k)|$. For every integer $c \neq 0$ whose absolute value is not a critical magnitude,*

$$\#\{x \in \mathbb{R} : f_{p,c}(x) = 0\} = 1 + 2 \#\{k \text{ odd} : |c| < \mu_k\}.$$

Proof. On $(k, k+1)$ exactly $k+1$ of the factors of φ are positive and $p-1-k$ are negative, so φ has sign $(-1)^{p-1-k} = (-1)^k$ there, p being odd; thus $v_k := \varphi(c_k)$ has sign $(-1)^k$, and consecutive critical values have opposite signs. By Proposition 15 the c_k are the only critical points, so φ is strictly

monotone on each of the p intervals $(-\infty, c_0), (c_0, c_1), \dots, (c_{p-2}, \infty)$, with $\varphi(x) \rightarrow \pm\infty$ as $x \rightarrow \pm\infty$. A level L is attained exactly once on each such interval on which it lies strictly between the endpoint values, and not at all otherwise.

Take first $c > 0$, so $L = -c < 0$. The leftmost interval always contributes, since $v_0 > 0 > L$. For $1 \leq j \leq p-2$ the interval (c_{j-1}, c_j) contributes iff L lies strictly between v_{j-1} and v_j ; these have opposite signs, so the span is (v_o, v_e) with o the odd and e the even index in $\{j-1, j\}$, and $L < 0 < v_e$ reduces the condition to $L > v_o$, i.e. $c < \mu_o$. The rightmost interval contributes iff $L > v_{p-2}$, i.e. $c < \mu_{p-2}$, and $p-2$ is odd. Now count. There are p monotonicity intervals: the leftmost, which always contributes; the rightmost; and the $p-2$ bounded ones. The leftmost is unconditional, and the remaining $p-1$ each carry the condition $|c| < \mu_k$ for one odd index k . Each odd k occurs for exactly two of them, namely (c_{k-1}, c_k) and (c_k, c_{k+1}) , the second of these being the rightmost interval when $k = p-2$. As there are $(p-1)/2$ odd indices in $\{0, \dots, p-2\}$, this accounts for $2 \cdot (p-1)/2 = p-1$ intervals, and $p = 1 + 2 \cdot \frac{p-1}{2}$ in total. The count follows.

For $c < 0$, the involution $x \mapsto (p-1) - x$ sends φ to $-\varphi$, hence $f_{p,c}((p-1) - x) = -f_{p,-c}(x)$; the two polynomials have the same number of real roots, so the count depends only on $|c|$. $\square$

Theorem 18 (Critical window). *Let $p \geq 5$ be prime, $m = (p-1)/2$, and let $c \in \mathbb{Z}$ satisfy $p \nmid c$ and*

$$\mu_{m-1} < |c| < \mu_{m-2}.$$

Then $f_{p,c}$ has exactly two non-real roots, and $\text{Gal}(f_{p,c}/\mathbb{Q}) = S_p$.

Proof. Proposition 14 gives $\mu_k = \mu_{p-2-k}$ and $\mu_0 > \mu_1 > \dots > \mu_{m-1} = \mu_m$. Since $p-2$ is odd, the involution $k \mapsto p-2-k$ reverses parity, so it matches each odd index with an even one; the odd indices therefore meet each mirror pair exactly once, and

$$\{\mu_k : k \text{ odd}\} = \{\mu_{m-1} < \mu_{m-2} < \dots < \mu_0\}$$

with each value occurring once. By Lemma 17, $f_{p,c}$ has $p-2j$ real roots, where $j = \#\{k \text{ odd} : \mu_k < |c|\}$; the stated window is precisely the condition $j = 1$. So $f_{p,c}$ has $p-2$ real roots and one pair of complex conjugate roots.

By Proposition 16, $f_{p,c}$ is irreducible of prime degree p , so $G = \text{Gal}(f_{p,c}/\mathbb{Q})$ is transitive and $p \mid |G|$; by Cauchy, G contains a p -cycle (or take Frob_p). Complex conjugation fixes the $p-2$ real roots and swaps the remaining pair, so G contains a transposition. Since p is prime, G is moreover primitive, and a primitive group containing a transposition is the full symmetric group; equivalently, a p -cycle together with a transposition generates S_p [1, Thm. 3.2]. Primality is used here as well: in composite degree a transitive group with a transposition need not be S_n [1, Rem. 3.3]. $\square$

Remark 19 (Why $c = 1$ is the hard case). The window is never reached by $c = \pm 1$. Indeed $\mu_k \geq |\varphi(k + \frac{1}{2})| \geq \min_j |\varphi(j - \frac{1}{2})| = (2m-1)!!(2m+1)!!/2^p$, which is $45/32 > 1$ at $p = 5$ and increases thereafter (Proposition 14); so $|c| = 1 < \mu_{m-1}$, $j = 0$, and f_p is totally real. That is Proposition 14 again, read off Lemma 17. The content of Theorem 18 is that the obstruction

is quantitative and narrow: the conjugation route is closed for $c = 1$ only because 1 lies below every critical magnitude, and it reopens as soon as $|c|$ crosses the smallest of them. The whole apparatus of §§2–7 is the price of the single value $c = 1$.

Remark 20 (Only the first window). Larger $|c|$ does not help. If j odd indices lie below $|c|$, complex conjugation is a product of j disjoint transpositions, of sign $(-1)^j$; Conrad’s criterion needs $j = 1$. At the top, $|c| > \mu_0$ leaves one real root and $(p-1)/2$ transpositions, an even permutation when $p \equiv 1 \pmod{4}$, which does not even exclude A_p . Theorem 18 is therefore sharp: the first window is the only one in which conjugation alone identifies the group. Outside it one falls back on Proposition 16.

The windows are wide. Writing N_p for the number of *positive* integers c in the window with $p \nmid c$ — each of which is matched by $-c$, since Lemma 17 depends only on $|c|$:

p	μ_{m-1}	μ_{m-2}	N_p
5	1.4187	3.6314	2
7	12.359	23.149	9
11	4804.9	7042.3	2034
13	171688	236529	59854
17	$5.3330 \cdot 10^8$	$6.7948 \cdot 10^8$	137 582 410
19	$4.3057 \cdot 10^{10}$	$5.3437 \cdot 10^{10}$	9 833 286 697

The window narrows — its ratio μ_{m-2}/μ_{m-1} falls from 2.56 at $p = 5$ to 1.24 at $p = 19$ — but μ_{m-1} itself grows like $p!/2^p$, so N_p grows superexponentially all the same: already at $p = 19$ there are some 9.8×10^9 admissible $c > 0$, and as many negative ones. For every prime p and every one of them, $\text{Gal}(f_{p,c}/\mathbb{Q}) = S_p$ is settled unconditionally, with no appeal to §7. The case $p = 5$ is worth recording: the certificate of Corollary 9 has no witness there, since $q = 3$ is the only candidate and $(38569/3) = +1$; but $c = 2$ and $c = 3$ lie in the window, so $\text{Gal}((x)_5+2) = \text{Gal}((x)_5+3) = S_5$ by Theorem 18 alone. The claims in the table, the root counts of Lemma 17, and the irreducibility of the corresponding $f_{p,c}$ are checked in `tools/verify_window.py`.

7. PERIODICITY OF THE DISCRIMINANT

The same identity that makes Frob_q fixed-point-free computes the Stickelberger symbol at a fixed odd prime $q < p$. Write $p = mq + r$ with $0 < r < q$, and set $g = x^q - x$, $B_r = (x)_r$. Then $\varphi \equiv g^m B_r \pmod{q}$. Since $r < q$, the factor B_r divides g ; write $g = B_r C_r$ with $C_r = \prod_{a=r}^{q-1} (x - a)$, and set

$$u_r := C_r B_r',$$

a polynomial of degree $q - 1$ depending only on r . (In particular $u_1 = x^{q-1} - 1$.) Differentiating and using $g' \equiv -1$ gives

$$\psi := g B_r' - m B_r = B_r(u_r - m).$$

The roots of B_r lie in $\mathbb{F}_q$, so by Proposition 1 they contribute 1 to the resultant. Multiplicativity together with $\text{Res}(g, f_p) = \prod_{a \in \mathbb{F}_q} f_p(a) = 1$ yields

$$(1) \quad \text{disc } f_p \equiv (-1)^{(p-1)/2} r^p \prod_{u_r(\beta)=m} (g(\beta)^m B_r(\beta) + 1)^{\mu_\beta} \pmod{q},$$

the product over the fibre $u_r^{-1}(m)$ in an algebraic closure, with multiplicity. (The leading coefficient of u_r is r , since B_r is monic of degree $r < q$.) The unbounded part of the discriminant has disappeared. What remains is periodic, and depends on p only through r , $m \bmod q$, and the values $g(\beta)^m$ at finitely many β of degree at most $q-1$.

Lemma 21 (Periodicity). *Let q be an odd prime and*

$$\Pi(q) = \text{lcm}\left(4, q^2, \text{lcm}_{1 \leq d \leq q-1}(q^d - 1)\right).$$

For odd primes $p, p' > q$, if $p \equiv p' \pmod{\Pi(q)}$ then $\text{disc } f_p \equiv \text{disc } f_{p'} \pmod{q}$, and in particular $(\text{disc } f_p / q) = (\text{disc } f_{p'} / q)$.

Proof. The right-hand side of (1) is defined for any odd integer $n = mq + r > q$ not divisible by q , via $f_n = (x)_n + 1$. The sign $(-1)^{(n-1)/2}$ has period 4. The fibre polynomial $u_r - m$ is determined by r and $m \bmod q$, hence by $n \bmod q^2$; this is the only source of the factor q^2 in $\Pi(q)$. The power r^n depends on $n \bmod (q-1)$, and $q-1$ divides the $d=1$ term of the inner lcm.

Each factor $g(\beta)^m B_r(\beta) + 1$ is an evaluation in a field. If $\beta \in \mathbb{F}_q$ then $g(\beta) = 0$, so the factor is 1. If $\beta \notin \mathbb{F}_q$ then $g(\beta) \neq 0$, because every root of g lies in $\mathbb{F}_q$, and $g(\beta)^m$ has period $\text{ord}(g(\beta))$ dividing $q^d - 1$ where $d = [\mathbb{F}_q(\beta) : \mathbb{F}_q] \leq q-1$.

If $n \equiv n' \pmod{\Pi(q)}$, these data agree. (For the multiplicative orders: $\gcd(q, q^d - 1) = 1$, so $n \equiv n'$ modulo q^2 and modulo $\text{lcm}_d(q^d - 1)$ implies $m \equiv m'$ modulo that lcm.) Restricting to primes yields the lemma. $\square$

Remark 22. The quadratic fibres are more rigid. Always $g(\beta) = \beta^q - \beta$ lies in $\ker \text{Tr}_{\mathbb{F}_{q^d}/\mathbb{F}_q}$. For $d=2$ the trace-zero hyperplane coincides with a multiplicative coset: $\text{Tr}(x) = x + x^q = 0$ if and only if $x^{q-1} = -1$ for $x \neq 0$. Thus if $\beta \in \mathbb{F}_{q^2} \setminus \mathbb{F}_q$ and $\gamma = g(\beta)$, then $\gamma^q = -\gamma$, so $\gamma^{q-1} = -1$. In particular $\text{ord}(\gamma)$ divides $2(q-1)$ and does not divide $q-1$. (For $d \geq 3$ the trace-zero set is a plain additive hyperplane of size q^{d-1} , with no multiplicative structure, and nothing of this kind carries over.) The factor $q^2 - 1$ in $\Pi(q)$ may therefore be replaced by $2(q-1)$.

Remark 23 (The period bound is sharp from $q=7$ on). The collapses at $q=3$ and $q=5$ are accidents. At $q=7$ the minimal period is exactly $\Pi(7) = 134064$: no even divisor of it is a period. What controls the sharp modulus is not the degrees of the fibres — those are already maximal, the part of $u_r - m$ coprime to g reaching degree $q-1$ for every $q \geq 5$ tested — but the exponent

$$E_q = \text{lcm}_\beta \text{ord}(g(\beta)),$$

the lcm being over all fibres. Measured,

$$E_3 = 4, \quad E_5 = 48, \quad E_7 = 5472, \quad E_{11} = 6.3 \times 10^{10}, \quad E_{13} = 5.1 \times 10^{13},$$

and the sharp period divides $\text{lcm}(4, q^2, E_q)$: with equality at $q=3$, and with index 2 at $q=5$ and $q=7$, the factor 2 coming from the parity coupling of m and r for odd p . Enumerating classes modulo such a modulus is therefore infeasible beyond $q=7$, and $\varepsilon_3, \varepsilon_5, \varepsilon_7$ are the only exact values this method will ever produce.

The bound is not always sharp: $\Pi(3) = 72$ while the actual period is 36. For $q = 3$ the fibres of $u_1 = x^2 - 1$ and $u_2 = 2(x + 1)^2$ can be read off by hand, and among odd p the value $\text{Res}(\psi, f_p)$ is constant on each class modulo 9.

Proposition 24. *Let $p > 3$ be prime. Then $(\text{disc } f_p/3) = -1$ if and only if*

$$p \equiv 7, 13, 17, 19, 23, 29 \pmod{36}.$$

Proof. Write $p = 3m + r$ with $r \in \{1, 2\}$ and $g = x^3 - x$. These are the fibres of $u_1 = x^2 - 1$ and $u_2 = 2(x + 1)^2$. The six values of $\psi = B_r(u_r - m)$ as $r \in \{1, 2\}$ and $m \pmod{3}$ are as follows. In each case $\text{Res}(\psi, f_p)$ is a constant $R \in \{1, 2\}$ on odd p , recorded with the corresponding class $p \pmod{9}$.

$r = 1$. Here $B_1 = x$, so $\psi = g - mx = x(x^2 - (m + 1))$.

- $m \equiv 0 \pmod{3}$: $\psi = g$. All roots lie in $\mathbb{F}_3$, so $f_p(\beta) = 1$ at every root and $R = 1$. Then $p \equiv 1 \pmod{9}$.
- $m \equiv 2 \pmod{3}$: $\psi = x^3$, so $R = f_p(0)^3 = 1$. Then $p \equiv 7 \pmod{9}$.
- $m \equiv 1 \pmod{3}$: $\psi = x(x^2 + 1)$. Let $\alpha^2 = -1$ in $\mathbb{F}_9$. Then $g(\alpha) = \alpha(\alpha^2 - 1) = \alpha$, so

$$R = (\alpha^{m+1} + 1)((-\alpha)^{m+1} + 1).$$

Odd $p = 3m + 1$ forces m even, hence $m + 1$ odd, and the product is $1 - (\alpha^2)^{m+1} = 1 - (-1) = 2$. Then $p \equiv 4 \pmod{9}$.

$r = 2$. Here $B_2 = x(x - 1)$ and $B'_2 = 2x - 1 = 2(x + 1)$ in $\mathbb{F}_3$, so

$$\psi = x(x - 1)(2(x + 1)^2 - m).$$

- $m \equiv 0 \pmod{3}$: $\psi = 2x(x - 1)(x + 1)^2$, all roots in $\mathbb{F}_3$, so $f_p(\beta) = 1$ at every root and $R = 2^p \equiv 2$. Then $p \equiv 2 \pmod{9}$.
- $m \equiv 2 \pmod{3}$: $\psi = 2x^2(x - 1)^2$, likewise $f_p(\beta) = 1$ at every root and $R = 2^p \equiv 2$. Then $p \equiv 8 \pmod{9}$.
- $m \equiv 1 \pmod{3}$: $\psi = 2x(x - 1)(x^2 + 2x + 2)$. The quadratic is irreducible ($\text{disc} \equiv 2$, not a square). For a root β one has $\beta^2 = \beta + 1$, $g(\beta) = \beta(\beta^2 - 1) = \beta^2$, and $B_2(\beta) = \beta^2 - \beta = 1$, so $R = 2^p N_{\mathbb{F}_9/\mathbb{F}_3}(1 + (\beta^2)^m)$. Now $(\beta^2)^2 = (\beta + 1)^2 = 2$, hence β^2 has order 4 in $\mathbb{F}_9^\times$. Odd $p = 3m + 2$ forces m odd, so $(\beta^2)^m \in \{\beta^2, -\beta^2\}$; both $1 + \beta^2 = \beta - 1$ and $1 - \beta^2 = -\beta$ have norm 2. Thus $R = 2^p \cdot 2 \equiv 1$. Then $p \equiv 5 \pmod{9}$.

The Kronecker symbol is $((-1)^{(p-1)/2}R/3)$. Since $(1/3) = 1$ and $(2/3) = -1$, this is $(-1)^{(p-1)/2}$ when $R = 1$ and the opposite when $R = 2$. Combining the class modulo 9 with the sign modulo 4 yields the six listed classes modulo 36, and no others among the units of $\mathbb{Z}/36\mathbb{Z}$. $\square$

The good set is stable under $p \mapsto -p$ (this fails already for $q = 5$). Dirichlet's theorem supplies infinitely many primes in each of the six classes, and Corollary 9 with $q = 3$ yields $\text{Gal}(f_p/\mathbb{Q}) = S_p$ on a set of Dirichlet density $\frac{1}{2}$. In particular:

Theorem 25. *There are infinitely many primes p with $\text{Gal}(f_p/\mathbb{Q}) = S_p$.*

This theorem is computer-free: it uses only Theorem 2 with Lemma 3 (the two finite checks in that proof are hand-verifiable), Proposition 24, and Dirichlet's theorem.

Each further prime q contributes its own periodic good set, of some Dirichlet density ε_q . The first three values are exact:

$$\varepsilon_3 = \frac{1}{2}, \quad \varepsilon_5 = \frac{11}{20}, \quad \varepsilon_7 = \frac{323}{648}.$$

(The last is 18088 of the 36288 units modulo 134064.) For $q = 5$, an inspection of the twenty fibres of u_r shows that the part coprime to g has degree 0, 2, or 4. Lemma 21 with $d \leq 4$, using Remark 22 for $d = 2$, yields period dividing $15600 = \text{lcm}(4, 5^2, 5^4 - 1)$. Evaluating (1) at one representative of each odd class modulo 15600 shows that the period divides 600, and no proper even divisor of 600 works. Among the 160 units of $\mathbb{Z}/600\mathbb{Z}$, 88 have symbol -1 , hence $\varepsilon_5 = \frac{11}{20}$. Combining with Proposition 24: of the 480 units of $\mathbb{Z}/1800\mathbb{Z}$, $372 = 240 + 264 - 132$ are good for $q = 3$ or $q = 5$ — the six classes modulo 36 lift to 240 units modulo 1800, the 88 classes modulo 600 lift to 264, and the intersection has 132 units. As $132 = 480 \cdot \frac{1}{2} \cdot \frac{11}{20}$, the two good sets are exactly independent on the units modulo 1800, which is not automatic since $\gcd(36, 600) = 12$.

The pattern does not continue. Evaluating (1) at every one of the 725,760 odd unit classes modulo $\text{lcm}(36, 600, 134064) = 3,351,600$ gives the exact joint law of the three symbols. The marginals come out at $\frac{1}{2}, \frac{11}{20}, \frac{323}{648}$, reproducing the values above on a modulus 25 times larger; the pair $\{3, 5\}$ is again exactly independent; but $\{3, 7\}$ is not. One has

$$\Pr[s_3 = -1, s_7 = -1] = \frac{787}{3024} = 0.26025\dots, \quad \varepsilon_3\varepsilon_7 = \frac{323}{1296} = 0.24922\dots,$$

so the good sets at $q = 3$ and $q = 7$ are positively correlated and the product formula $1 - \prod(1 - \varepsilon_q) = \frac{511}{576}$ is wrong.

Theorem 26. *The set of primes p with $(\text{disc } f_p/3) = -1$ has Dirichlet density $\frac{1}{2}$; adjoining $q = 5$ gives $\frac{31}{40}$; adjoining $q = 7$ gives*

$$\frac{7117}{8064} = 0.88256\dots$$

On each of these sets, $\text{Gal}(f_p/\mathbb{Q}) = S_p$.

The last of these is also the last one obtainable this way, for the reason given in Remark 23.

The least-witness statistics over the 664577 primes $5 \leq p < 10^7$ exhibit the joint law rather than merely the marginals. For $q \in \{3, 5, 7\}$ the least witness is a function of p modulo 3351600, and it is the same function that the class enumeration tabulates; so what follows is not a second evaluation of the symbols but a check that the primes sample that class function faithfully, and a demonstration that the correlation can be read off the witness list given only $\varepsilon_3, \varepsilon_5, \varepsilon_7$. Writing P_q for the proportion of primes whose least witness is q :

	$q = 3$	$q = 5$	$q = 7$
observed	0.50003	0.27483	0.10767
joint law	$\frac{1}{2} = 0.50000$	$\frac{11}{40} = 0.27500$	$\frac{4337}{40320} = 0.10756$
discrepancy	0.05σ	0.31σ	0.27σ

Had the three symbols been independent, P_7 would be $(1 - \varepsilon_3)(1 - \varepsilon_5)\varepsilon_7 = 0.11215$, which the sample excludes at 11.8σ . That the symbols are correlated is not news — Theorem 26 proves it — but the exclusion shows the correlation is legible in the witness list itself, at a sample size where the two models are far apart; below 10^5 they differ by only about 1.5σ . Likewise $P_3 + P_5 + P_7 = 0.882525$ against the predicted $\frac{7117}{8064} = 0.882564$, a discrepancy of 0.1σ .

Remark 27 (Ramification is rare, and computable). The same computation exhibits a set of unit classes of density $\frac{1}{324}$ on which $s_7 \equiv 0$: every prime in them has $7 \mid \text{disc } f_p$. No such classes occur at $q = 3$ or $q = 5$. Ramification of a fixed ℓ is therefore periodic in p like the symbol itself, and — unlike the symbol — its density can be computed without knowing the period. Write δ_ℓ for the density, among odd integers n coprime to ℓ , of those with $\ell \mid \text{disc } f_n$; this is the quantity the fibres compute. It majorises the density among primes, which is smaller because primes avoid the classes divisible by the other prime factors of the period: at $\ell = 7$ the prime density is $\frac{1}{324}$ against $\delta_7 = \frac{50}{3591}$. It is in turn larger than the density of the simple ramification $v_\ell = 1$ that the transposition certificate needs. For the negative conclusion below that is the favourable direction.

A factor of the product in (1) vanishes exactly when $g(\beta)^m = -1/B_r(\beta)$, and only β of degree ≥ 2 can contribute, since $\beta \in \mathbb{F}_\ell$ gives $g(\beta) = 0$ and a factor 1, while $B_r(\beta) \neq 0$ because B_r splits over $\mathbb{F}_\ell$. The exponent is not free: it is pinned modulo ℓ by the fibre $u_r(\beta) = m$ and modulo 2 by p odd, so m runs over a single class M modulo 2ℓ and $g(\beta)^m$ over the coset $\gamma^M \langle \gamma^{2\ell} \rangle$, $\gamma = g(\beta)$. Writing $o = \text{ord } \gamma$ and $e = \gcd(o, 2\ell)$, the group $\langle \gamma \rangle$ is the unique subgroup of its order in the cyclic group $\mathbb{F}_{\ell^d}^\times$, so $\langle \gamma^{2\ell} \rangle = \langle \gamma^e \rangle$ and membership is the single test $x^{o/e} = 1$. The vanishing class then has conditional density e/o , whence

$$\delta_\ell \leq \frac{1}{\ell(\ell-1)} \sum \frac{e}{o},$$

a density over odd n coprime to ℓ , the sum running over those r , m_0 and irreducible factors of $u_r - m_0$ of degree ≥ 2 for which $(-B_r(\beta)^{-1} \gamma^{-M})^{o/e} = 1$. This is a finite computation for each ℓ , independent of the period — which matters, since by Remark 23 the period is out of reach for $\ell \geq 11$.

The inequality is the honest statement, because ramification is the event that *some* factor vanishes while the right-hand side adds one term per factor. Distinct fibres never overlap: two of them differ either in $r = p \bmod \ell$ or in $m \bmod \ell$. So the only possible overcount is between two factors of one fibre (r, m_0) , and equality holds as soon as each fibre contributes at most one passing factor. That is the case for $\ell = 7$ and $\ell = 11$, where the passing factors lie in 12 and 32 distinct fibres respectively; from $\ell = 13$ on some fibres carry two or three, and the displayed sum is then an upper bound, overshooting by $O(\delta_\ell^2)$.

At $\ell = 7$, where equality holds, the formula gives $\delta_7 = \frac{50}{3591}$, agreeing exactly with an enumeration over the full period 134064. At $\ell = 3$ and $\ell = 5$ the sum is empty, so 3 and 5 *never divide* $\text{disc } f_p$, for any p ; for $\ell = 3$ that is already visible in Proposition 24, where $R \in \{1, 2\}$.

These densities are small. Only $\ell = 3$ and $\ell = 5$ are excluded outright, by the empty sum; at $\ell = 11, 13, 19, 31$ and 43 the density is merely small — of order 10^{-3} — and no prime below 10^5 happens to be ramified there, which is a statement about that range and not about all p . Summed over odd $\ell \leq 199$, the densities come to 0.007. The cause is the same smallness of $\langle g(\beta) \rangle$ as in Remark 22: a factor vanishes only when a prescribed target lands in a coset of that subgroup. In particular the ramification classes do not cover: the transposition certificate of §5 fires through ramified primes as large as 669577, not through small ℓ , and such primes are not found by trial division once $\text{disc } f_p$ has $\asymp p^2 \log p$ digits.

8. WHAT REMAINS

Write $\text{disc } f_p = (-1)^{(p-1)/2} \text{Res}(\varphi', \varphi + 1)$. Reducing modulo p (where $\varphi' \equiv -1$) gives the congruence

$$\text{disc } f_p \equiv (-1)^{(p+1)/2} \pmod{p},$$

always a quadratic residue, as forced by the even p -cycle Frob_p . This is a consistency check, not a constraint on witnesses $q < p$.

The identification $\text{Gal}(f_p/\mathbb{Q}) = S_p$ for every odd prime p is a covering question, of two different kinds. For a *fixed* finite set of q , periodicity plus Dirichlet makes the covered set a union of arithmetic progressions, of a density given by the joint law of the symbols (Theorem 26; not the product $1 - \prod(1 - \varepsilon_q)$, which already fails for $\{3, 7\}$). A finite list cannot cover every prime: the data already refutes covering by $q \leq 71$, since $p = 9683099$ has least witness 73. The least witness appears unbounded, so all- p is unlikely to fall to a fixed finite set of congruences.

Density 1 is a class-density statement: if $\varepsilon_q \geq c > 0$ then $\sum \varepsilon_q = \infty$, and the uncovered density tends to 0 as more q are adjoined. For each fixed Q this is rigorous given periodicity and Dirichlet, once the joint law is computed. It does not imply that every prime is covered. For a given p the available witnesses are the $q < p$, so $Q \sim p$ and the modulus Π up to that Q is far larger than p ; equidistribution modulo that Π has no bearing on whether that particular p sits in a bad class.

The barrier is therefore a uniform lower bound $\varepsilon_q \geq c > 0$ on the exact rationals of §7. Each q has $q(q-1)$ fibres (r, m_0) . On a fibre the residual R of (1) may be constant in m . The symbol is $\chi_q((-1)^{(p-1)/2}R)$, so a constant- R fibre is constant as a symbol if and only if $\chi_q(-1) = +1$ (equivalently $q \equiv 1 \pmod{4}$); if $\chi_q(-1) = -1$ the sign still flips it. This is exact at the three q whose periods can be enumerated:

- $q = 3$: all 6 fibres have constant R , none have constant symbol, and $\varepsilon_3 = \frac{1}{2}$.
- $q = 5$: of 20 fibres, 10 have constant R and all 10 have constant symbol. Six are always -1 and four always $+1$; the remaining ten are balanced. Hence $\varepsilon_5 = (6 + 5)/20 = \frac{11}{20}$. Six of the ten constants are the fibres on which $u_r - m_0$ splits over $\mathbb{F}_5$, and the symbol is then $\chi_5(r)$: $(r, m_0) = (1, 0), (1, 4), (4, 0), (4, 4)$ with $\chi_5(r) = +1$, and $(2, 0), (3, 4)$ with $\chi_5(r) = -1$. The excess of two always- -1 fibres in twenty is the $+\frac{1}{20}$.

- $q = 7$: of 42 fibres, 14 have constant R and none have constant symbol. Among unramified unit classes one has exactly $\frac{1}{2}$; among all unit classes, $\varepsilon_7 = \frac{323}{648}$, the deficit $\frac{1}{648}$ being ramification (Remark 27).

A character-sum bound over $\langle g(\beta) \rangle$ cannot supply $\varepsilon_q \geq c$: already for quadratic fibres Remark 22 gives $|\langle \gamma \rangle| \leq 2(q-1)$, and a fibre whose symbol is identically $+1$ is not cancelled by mixing on other coordinates. What would suffice is coarser: writing, at these three q ,

$$\varepsilon_q = \frac{\#\{\text{always } -\} + \frac{1}{2} \#\{\text{balanced}\}}{q(q-1)}$$

up to ramification, a uniform $c > 0$ follows as soon as a positive proportion of the $q(q-1)$ fibres are non-degenerate. That is a statement about the maps u_r , exact at $q = 3, 5, 7$. We do not claim the trichotomy $\{0, \frac{1}{2}, 1\}$ beyond those q .

The good sets are explicit congruence classes, computed once per q . (The $q = 3$ symmetry $p \mapsto -p$ already fails for $q = 5$, so there is no functional equation of that form across q .) Any analytic argument that disc f_p is a non-residue at some $q < p$ need only treat primes in the complementary classes $1, 5, 11, 25, 31, 35 \pmod{36}$, a density- $\frac{1}{2}$ family, rather than all p . (The cases $p = 3, 5$ are settled by other witnesses, as above.) The covering, and the identification for every p , have been verified for all odd primes $p < 10^7$: 664577 rows. Each row is independently checkable by one resultant $\text{Res}(f'_p, f_p)$ and one Legendre symbol over $\mathbb{F}_q$, and that is how a sample of them is audited; a full audit of the list instead uses the $O(q^2 \log p)$ evaluation of §7, the two having been compared at every scale up to 10^7 .

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